Inference as Consolidation: Continual Learning with No Offline Phase by Replay in the Silent Degrees of Freedom

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Abstract

An agent that keeps learning in deployment may not be able to afford dedicated consolidation downtime, yet replay-based continual learning almost always relies on offline rehearsal phases or on interleaving replayed samples with the incoming stream. Brains face the same constraint and solve part of it with *local sleep*: brief, use-dependent off-periods of individual circuits during wakefulness. We ask whether a network trained by local, biologically constrained rules can consolidate *during inference*, with no offline phase at all. We introduce (i) an *isolation* rule that confines replay updates to hidden synapses invisible to the current input under k-winner-take-all dynamics — exact for silent presynaptic and suppressed units, near-exact elsewhere by measurement — (ii) a *refractory rotation* rule under which units that have just fired sit out the next competition, widening the isolated channel, and (iii) internal control signals (homeostatic pressure triggering replay bursts, a relative-novelty gate). The construction inverts the usual direction of non-interfering continual learning: the hidden computation on the current input, under its suppression mask, is held invariant — exactly on the proven channels, and for all but 0.3% of waking hidden codes per update elsewhere — while past memories are consolidated through degrees of freedom the current batch leaves unused. On class-incremental split-MNIST the system reaches $91.6 \pm 0.3\%$ with no offline phase, above the best offline-rehearsal schedule ($89.2 \pm 1.7\%$), at $2.4\times$ its replay samples; about $1.22\times$ of that budget, eight-sample micro-batches still reach $91.1 \pm 0.4\%$. Rotation carries the gain (rotation with unmasked replay reaches $91.5 \pm 0.3\%$); isolation adds 0.1–0.3 points at replay batches of 16 to 4 and the guarantee, 2.6 without rotation, and at two-sample batches removes a collapse mode. The ordering transfers to split CIFAR-10, and every reported number is measured on held-out training data never used for any decision. A drive-referenced controller for the rule's synaptic decay replaces per-dataset tuning with one dataset-independent ratio target and, with locally learned skip synapses, holds a five-hidden-layer stack at the two-layer level where the tuned fixed decay without skips collapses.

1 Introduction

Foundation-model and embodied agents are increasingly expected to keep learning after deployment, from streams that are non-stationary and never task-labelled. The dominant tool against forgetting is replay from an episodic memory, and nearly every replay-based method consumes it in one of two ways: in a dedicated offline phase — a "night" during which the agent stops acting and rehearses — or by interleaving replayed samples with the live stream. Both cost an agent. An offline phase is downtime during which the system is unavailable or silently changing; interleaved replay perturbs

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the computation currently serving the input, and grows fragile as replay batches shrink toward what an online budget allows.

Biology suggests a third option. Sleep-like off-periods occur *locally*, in individual cortical circuits of awake animals [Vyazovskiy et al., 2011], and inducing such ON/OFF alternation in one hemisphere of awake mice discharges sleep pressure and rescues memory consolidation, whereas an equal tonic reduction of firing does not [Driessen et al., 2026]. Some core functions of sleep can thus be discharged locally during wakefulness. We take this as an algorithmic motivation — not as a claim about where biological consolidation occurs — to ask whether the computational degrees of freedom a learner is not using can host consolidation while it acts, and what that costs.

We study the question in a deliberately constrained learner — a two-hidden-layer network trained end to end by local, biologically motivated rules (no weight transport, bounded burst-like error channel, Dale's law, sparse population codes, sparse distance-dependent wiring, single forward–feedback sweep) — because those constraints expose exactly which degrees of freedom the current input does and does not use. The contributions are:

- **Isolated replay during inference.** A synapse may take a replay update only if its presynaptic unit is silent for the current input or its postsynaptic unit is asleep. Under k-WTA dynamics this leaves the hidden computation exactly invariant for silent-presynaptic and suppressed units (Propositions 1–2); we measure how nearly it holds elsewhere and end to end. With rotation on, replay micro-batches of 8–16 samples suffice, where unmasked replay without rotation and offline schedules degrade; isolation adds 0.1–0.3 points there (2.6 without rotation), the guarantee, and at two-sample batches the absence of a collapse mode; replay confined to the readout falls to 42%: hidden-layer replay updates are essential.
- **Refractory rotation.** Units that fired on the current batch sit out the next competition. This alternation — the algorithmic counterpart of the ON/OFF requirement in Driessen et al. [2026] — widens the consolidable channel from ≈ 0.25 to 0.53 of the replay-active units and is worth $+3.6$ sequential points on top of isolation ($+6.1$ alone); a random alternation of matched size recovers most of that, use-dependence the remaining 0.6 sequential and 2.0 static points; a tonic reduction, like the biological control, loses on both axes.
- **Internal triggers** (schedulers for sparse replay budgets; the headline system replays at every batch). Replay is triggered by a unit-level homeostatic pressure; the tested surprise trigger fails during wake, whereas offline rehearsal is best timed by novelty in our sweeps. A relative-novelty gate on rotation is scale-free where absolute thresholds fail across the regimes considered here.
- **One ratio target instead of per-dataset decay tuning, and depth.** A drive-referenced controller for the rule's synaptic decay removes its last per-dataset constant and, with locally learned skip synapses, holds a five-hidden-layer stack at the two-layer level on both axes where the tuned fixed decay without skips collapses to chance.

Everything is evaluated on two axes at once — the class-incremental stream and the identical learner on the i.i.d. shuffle of the same data — because consolidation during inference must not damage ordinary learning. The scale is small (MLPs on MNIST/CIFAR variants); the object of study is the mechanism, whose assumptions (sparse codes, a local error signal) are the ones a larger system would have to satisfy to inherit the guarantee.

2 Related work

Sleep-inspired consolidation. The Bazhenov line implements sleep as a global offline phase driven by noise under local plasticity [Tadros et al., 2022, Bazhenov et al., 2026, Golden et al., 2022, Kubo et al., 2025]; wake–sleep frameworks with NREM/REM stages [Sorrenti et al., 2025, Robinson et al., 2022] and systems such as SIESTA [Harun et al., 2023] likewise schedule consolidation offline, and SESLR [Lin et al., 2026] appends a noise-enhanced sleep phase (a frozen-extractor spiking classifier training only on its buffer) to correct online learning's recency bias. In all of these sleep is global, offline and clocked; none consolidates in currently unused units during inference, and the bias correction such a phase delivers is what our plastic readout receives continuously from isolated replay.

Gating and history-dependent competition. Context-dependent gating activates a subnetwork per task from task identity [Masse et al., 2018]; learned gates [Tilley et al., 2023], dendritic context [Iyer et al., 2022], task-free sparsification [Lässig et al., 2023] and recovered functional masks [McKee et al., 2026] derive the gate from the input. Refractory competition has classical precedents [Kaski and Kohonen, 1994, Maeda and Miyajima, 1999], and spiking continual learners raise firing thresholds with use [Shen et al., 2024]. Our rotation differs in purpose: recently active units are benched for one competition to enlarge a *replay channel*, not to allocate resources across tasks.

Null-space and parameter isolation. A mature line protects *previous* tasks: gradient or activation-subspace projections [Farajtabar et al., 2020, Saha et al., 2021], feature-covariance null spaces [Wang et al., 2021], k-winner sparsity with such projections [Abbasi et al., 2022], frozen subnetworks [Golkar et al., 2019, Mallya and Lazebnik, 2018]. We invert the protected direction: the ongoing computation is held invariant while old-memory replay is written into degrees of freedom the current sparse support exposes as invisible — no stored bases or task masks, and the "null space" changes every batch for free. McKee et al. [2026] and Bricken et al. [2023] note that optimiser state can break such guarantees; ours is mask-confined. Complementary-learning-system methods [Arani et al., 2022, Pham et al., 2021] raise the value of each replayed sample but still replay at every step; learned replay scheduling [Klasson et al., 2023] is related to our trigger study.

3 Method

Setting. A stream of labelled examples is presented in T contiguous tasks with disjoint class sets; no task identity is available at test time; a single shared C-way readout is evaluated on all classes after the last task (class-incremental learning). We report final accuracy A_{seq}, forgetting F, the accuracy A_{iid} of the identical learner and budget on the i.i.d. shuffle (the *static* axis), the replay cost R in samples, and an arithmetic-energy proxy E; the learner is confined to the cortical constraint set $\mathcal{B}$ above and to an episodic memory of K raw examples with reservoir writing [Vitter, 1985]. The question is whether the offline term of consolidation can be removed entirely without losing A_{seq}, and at what price in A_{iid} and R.

Base learner. Layer activities are $a^0 = x$ and, for $\ell = 1, \ldots, L$,

$$z^\ell = W^\ell a^{\ell-1} + b^\ell, \qquad a^\ell = \Phi_k\big(\sigma(z^\ell) \odot (1 - s^\ell)\big), \tag{1}$$

where σ is a rectifier, $s^\ell \in \{0,1\}^{n_\ell}$ the suppression mask of the rotation rule below, and Φ_k keeps the k largest entries of each row (k-WTA); a per-unit gain exists in the implementation for homeostatic scaling but is fixed at 1 throughout. Layers $1, \ldots, L{-}1$ are hidden; layer L is a linear C-way readout whose pre-activation z^L is the output, scored by squared error against the one-hot target y. Errors come from one top-down sweep through learned feedback weights $B^\ell \in \mathbb{R}^{n_{\ell+1} \times n_\ell}$ (no weight transport),

$$\varepsilon^L = y - z^L, \qquad \varepsilon^\ell = \sigma'(z^\ell) \odot \Big((B^\ell)^\top \big[\kappa \tanh(\varepsilon^{\ell+1}/\kappa) \odot \mathbf{1}(a^{\ell+1} > 0)\big]\Big), \tag{2}$$

a bounded ($|\varepsilon_i^\ell| \le \kappa \sum_j |B_{ji}^\ell|$ per component), event-gated burst signal in the spirit of predictive-coding and burst-dependent learners [Whittington and Bogacz, 2017, Payeur et al., 2021]. Forward and feedback weights follow the same local product with a shared decay λ (Kolen–Pollack alignment [Kolen and Pollack, 1994, Akrout et al., 2019]), $\Delta W^\ell \propto \varepsilon^\ell (a^{\ell-1})^\top - \lambda W^\ell$, $\Delta B^{\ell-1} \propto \varepsilon^\ell (a^{\ell-1})^\top - \lambda B^{\ell-1}$, under Dale's law with sign-concordant feedback. The optimiser is heavy-ball SGD — one per-synapse velocity, no second-moment state (Section 4: masked Adam ties it, momentum-free SGD does not serve inside the consolidation loop). The base learner costs 1.2 points against a dense backpropagation MLP on i.i.d. MNIST at $4\times$ fewer synaptic operations (Appendix C).

Isolated replay. Let $X = \{x_b\}_{b=1}^m$ be the current waking batch, $a_{i,b}^\ell$ the activity of unit i on sample b, and $\mathcal{A}^\ell(X) = \{i : \exists b,\ a_{i,b}^\ell \neq 0\}$ the awake set. During the same forward step, a replay micro-batch drawn from the buffer is applied through the synapse mask

$$M_{ij}^\ell = \mathbf{1}\big[\, j \notin \mathcal{A}^{\ell-1}(X) \ \vee\ i \notin \mathcal{A}^\ell(X) \,\big], \tag{3}$$

i.e. iff the presynaptic unit is silent for the current input or the postsynaptic unit is asleep. The mask applies to the *entire* replay-induced parameter change, not only to the data gradient: with heavy-ball velocity v ($\mu = 0.9$), the local update direction $G = \varepsilon^\ell (a^{\ell-1})^\top$ and the per-step decay coefficient λ, a replay step is

$$v \leftarrow M \odot (\mu v + G) + (1 - M) \odot v, \qquad W \leftarrow W + \eta\, M \odot v - \lambda\, M \odot W, \tag{4}$$

so the velocity is frozen (not zeroed) outside the mask, the decay is confined to it, and biases follow the unit mask $\mathbf{1}[i \notin \mathcal{A}^\ell(X)]$; unmasked momentum would re-inject the update into awake synapses [McKee et al., 2026, Bricken et al., 2023]. The readout row and its bias stay plastic, so old classes remain calibrated; the guarantees below concern the hidden computation. Within one step the order is: forward pass under the current suppression, unmasked waking update, awake sets from that pass, masked replay on the updated weights; replay is inferred with the suppression removed (a refractory unit may fire on replay and learn from it) and a burst reuses one mask (Algorithm 1).

Proposition 1 (Pre-silent updates are exactly invisible)**.** *Fix X and let $\widehat{W}^\ell = W^\ell + \Delta^\ell$ with $\Delta^\ell_{ij} = 0$ whenever $j \in \mathcal{A}^{\ell-1}(X)$. Then every hidden activity on every sample of X is unchanged for any magnitude of Δ; if the readout row satisfies the same condition, or is held fixed, the network output is unchanged as well.*

Proposition 2 (Post-asleep updates are invisible under a margin)**.** *Let Δ^ℓ and Δb^ℓ additionally move synapses and biases of units $i \notin \mathcal{A}^\ell(X)$ with active presynaptic partners, and write $\delta_{i,b} = \sum_j \Delta^\ell_{ij} a^{\ell-1}_{j,b} + \Delta b^\ell_i$. If for every such unit and every sample $\sigma(z^\ell_{i,b} + \delta_{i,b}) < \tau^\ell_{k,b}$, where $\tau^\ell_{k,b}$ is the k-th largest value entering Φ_k on sample b (strict, so a tie cannot promote the unit; when fewer than k entries are positive, $\tau^\ell_{k,b} = 0$ and the condition reads $\sigma(\cdot) = 0$), the hidden activities on X are unchanged.*

Corollary 1 (Suppressed units are exact for any magnitude)**.** *If $s^\ell_i = 1$ in (1), $a^\ell_{i,b} = 0$ for every sample and Proposition 2 holds with no margin condition.*

Proofs are in Appendix A. The default system does not enforce the margin, so we measured it over all 18,760 isolated replay updates of the record run (three seeds): the update altered the top hidden code of 0.32% of waking samples (0.29–0.36) and a waking prediction for 0.33% (almost all through the plastic readout; 0.001% with it isolated), violating the margin for $\sim 10^{-4}$ of asleep units per step, with a mean maximal logit change of 0.02 (0.001 isolated) on the one-hot scale. The isolation is exact for silent-presynaptic and suppressed units and near-exact elsewhere for the hidden code of the *current* batch under its training-time suppression mask (evaluation runs with the mask off); later inputs are the ablation's business (Table 1), and Corollary 1 is why rotation makes the isolation robust.

Refractory rotation. After batch X_t, set $s^\ell_i(t{+}1) = \mathbf{1}[i \in \mathcal{A}^\ell(X_t)]$: every unit that fired sits out the next competition and is therefore asleep — consolidable, and exactly so — for X_{t+1}. Rotation widens the isolated channel $\rho_\ell = |\tilde{\mathcal{A}}^\ell \setminus \mathcal{A}^\ell(X)|/|\tilde{\mathcal{A}}^\ell|$ — the fraction of the units $\tilde{\mathcal{A}}^\ell$ activated by the replay micro-batch that are asleep for the current input — from 0.2–0.3 to ≈ 0.53, at the price of running inference on the second-best coalition.

Internal control (Figure 1b). *When to replay:* each unit integrates its own use, $S_i \leftarrow S_i + \bar{a}_i(X_t)$ with $\bar{a}_i(X_t)$ the fraction of the batch's samples on which it fired — a unit-level Process S [Borbély, 1982] — and a burst of n_b replay micro-batches fires when the mean pressure over the currently asleep units of all hidden layers exceeds θ; consolidation resets the pressure of the units that received it ($S_i \leftarrow 0$). *When to rotate:* with fast and slow exponential averages of the batch-mean squared output error e_t, $\bar{e}_f$ (α_f=0.1) and $\bar{e}_s$ (α_s=0.002), rotation runs while $\bar{e}_f \geq \beta\, \bar{e}_s$ ($\beta \approx$ 1.1–1.5) or while $\bar{e}_f \geq \gamma e_0$ with e_0 the chance-level error of the first 20 batches ($\gamma \approx 0.5$): the stream is new relative to the learner's own recent history, or is not yet mastered. The detector is a standard drift monitor [Ross et al., 2012], loosely inspired by cholinergic novelty signalling [Hasselmo, 2006]; what is new is what it gates, and a single absolute threshold cannot serve across the regimes considered here (a converged ten-class i.i.d. error exceeds any level a two-class task dips under). At the activity fraction used throughout the gate is nearly neutral, and the default system runs rotation always on (Section 4).

Protocol. Split-MNIST (five tasks of two classes, five epochs per task) and split CIFAR-10 (same protocol, 32×32 luminance), each with its i.i.d. counterpart; a third regime feeds CIFAR-10 through a frozen V1-like patch front-end (1024-D, linear probe $\approx 49\%$); split CIFAR-100 appears in Section 5

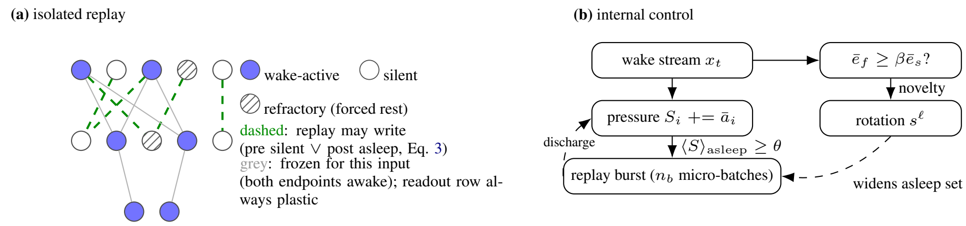

Figure 1: **Mechanism.** (a) During each waking batch, replay from the episodic buffer is written only into hidden synapses invisible to the current input — exactly so for silent presynaptic and suppressed units; refractory rotation forces recently used units into the consolidable set. (b) Unit-level homeostatic pressure triggers replay bursts; a relative-novelty gate decides when rotation runs. The default system runs rotation always on with one micro-batch per waking batch; the pressure trigger and the novelty gate are schedulers for sparse budgets and lower activity fractions.

for mechanism only. Architecture d–512–256–C, k-WTA 10%, 30% distance-dependent connectivity, 80% excitatory units; waking batches of 16; buffer K=1000 (K=200 in Appendix G). Baselines: BP with experience replay (BP+ER) at equal K; the offline "night" (20 replay batches of 256 at $3\times$ learning rate after each waking epoch, the best offline schedule in our sweep) on its preferred narrower core and on our substrate; interleaved ER for the local learner; no buffer. Three seeds (six on headline rows), mean $\pm$ s.d. The official test sets served as the development set throughout the project; every number reported here is measured on a stratified tenth of the training data held out from training and never used for any decision, with the remaining nine tenths as the stream (a second, independent tenth replicates the headline rows, Appendix L). Forgetting F is the mean drop of each earlier task's accuracy from its own end to the end of training (per-task matrix in Appendix L). Evaluation runs with the suppression mask off (k-WTA only), so test order and batch size cannot affect it. Energy is an idealised 45-nm FP32 arithmetic proxy [Horowitz, 2014] (0.9 pJ per accumulate, one per synaptic event, against 4.6 pJ per multiply–accumulate) after rate-to-spike conversion with training-calibrated thresholds [Rueckauer et al., 2017], excluding data movement and membrane updates. Further protocol details are in Appendix D.

4 Results

Local sleep replaces the night. On split-MNIST at K=1000, refractory local sleep with one isolated replay micro-batch beside every waking batch reaches $\mathbf{91.6 \pm 0.3}\%$ (six seeds) with no offline phase — above the best offline night ($89.2 \pm 1.7\%$ on its preferred narrow substrate; 89.0 ± 3.5, highly variable across seeds, on ours; paired margins $+2.5$ $[+1.3, +4.8]$ and $+2.6$ $[+0.2, +5.2]$), above unmasked interleaved ER at the same budget (87.5 ± 0.8 under Adam, 85.4 ± 3.5 under SGD) and above BP+ER (88.8 ± 0.3; $+2.8$ $[+2.5, +3.2]$). A night added on top adds 0.8 (92.4 ± 0.3). The headline configuration replays $2.4\times$ the night's samples; around $1.22\times$ of the night's samples (22% more) the system still reaches 91.1–91.2 (eight-sample micro-batches every batch, $39\times$ the night's updates, or 16-sample ones every second batch, $19.5\times$; Figure 2, Appendix D). On a second, independent held-out split the ordering holds (rotation 91.8 ± 0.4, BP+ER 88.5 ± 0.7, unmasked ER 80.0 ± 14.1, the narrow night 89.3 ± 0.6), with one exception worth stating: the same-substrate night, bimodal on the first split, reaches 91.5 ± 0.5 on the second and trails by only 0.3 there (Appendix L).

What rotation and isolation each buy. Figure 2 (left): the full system is flat in replay batch size from 256 down to 8 ($91.6 \pm 0.5 \to 91.1 \pm 0.4$) and loses 1.9 at 4 ($89.7 \pm 0.7$); at batch 16 unmasked replay without rotation sits at 85.4 ± 3.5, isolation without rotation at 88.0 ± 0.7, and the offline night falls from 89.0 to 85.8 ± 3.0 when its batches shrink to 16. The fourth cell — rotation with unmasked replay — reaches 91.5 ± 0.3 at batch 16, 90.9 ± 0.1 at 8 and 89.3 ± 0.7 at 4: the factors are strongly sub-additive, rotation is the larger one ($+6.1$ alone, $+3.6$ on top of isolation), and isolation's seed-paired margin is 0.1 $[-0.3, +0.5]$ at batch 16, 0.3 $[-0.2, +0.6]$ at 8 and 0.3 $[-0.5, +1.2]$ at 4 — within seed noise at every size — with 0.4 on the static axis (94.1 vs. 93.7). At two-sample batches both lose heavily, but isolated replay degrades gracefully (79.0 ± 3.2, every seed within 75–84) where unmasked replay collapses in three of six seeds (Appendix K): isolation's accuracy value is the removal of a failure mode, not a margin. What it buys at every batch size is the guarantee

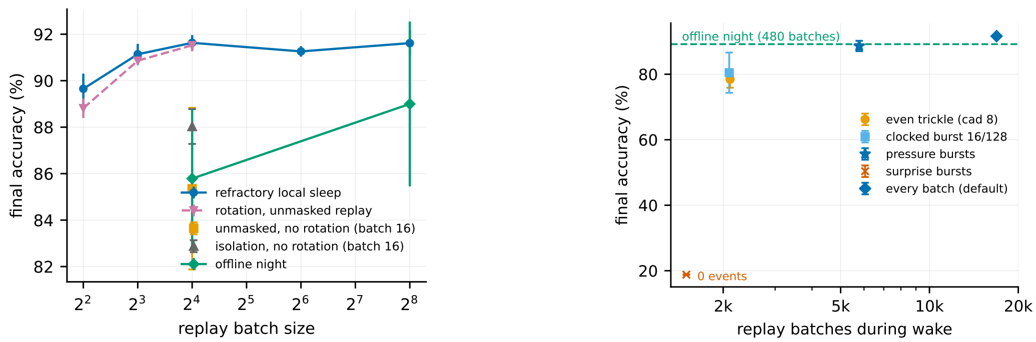

Figure 2: **Left (isolation):** split-MNIST accuracy against replay batch size, the number of updates fixed within each series (one per waking batch; 20 per epoch for the night, whose samples therefore fall with its batch): the full system is flat down to 8 samples and loses 1.9 at 4 (12.7 at 2, Appendix K); rotation with unmasked replay (dashed) trails it by 0.1 to 0.3, within seed noise, and collapses in half the seeds at batch 2; without rotation, unmasked and isolated replay and the offline night all degrade. **Right (timing):** accuracy against the number of waking replay events: trickles and clocked bursts fail alike, pressure-triggered bursts (star) keep most of the accuracy at a third of the events, surprise triggering fires no events in this regime.

that, for the current input under its suppression mask, replay leaves the hidden activities unchanged — exactly on the proven channels, within the margin elsewhere; without it replay micro-batches can perturb the live coalition and the outcome rests on the rotation's tolerance of that perturbation.

Component prices (Table 1). Removing rotation costs 3.6 sequential points (88.0 ± 0.7: the same mask on natural silence alone), removing both 6.2 with a twelve-fold larger seed deviation (85.4 ± 3.5), removing isolation alone 0.1 (above), and removing replay everything (19.4, the forgetting floor). Replay confined to the readout row, hidden layers frozen for replay, falls to 41.7 ± 10.9 (ten classes) and costs 11 static points: readout-only replay cannot account for the full system's accuracy, and hidden-layer replay updates are essential to it. A random suppression of matched size recovers most of rotation's gain (91.0 ± 0.8) but trails it by 0.6 sequential and 2.0 static points with $1.4\times$ the forgetting: alternating the coalition carries most of the effect, resting the recent winners the rest. Rotation is the only statically costly component, 1.2–1.7 points at this activity fraction (94.1 against 95.3–95.8 without it; isolation adds 0.4). The unmasked control is not handicapped by the replay step (3η against a waking step of $\eta/16$): over replay gains $\{1/16, 1/4, 1, 3, 10\}$ both curves peak at the default (91.6 masked, 85.4 unmasked), gain 10 destabilises both, and the rotation-free unmasked control never closes the gap; the sweep bounds that control, it does not estimate isolation's own contribution (the square above does; Appendix I). *Mirror isolation* (waking updates barred from synapses that served the last replay batch, replay unmasked) reaches only 78.0 ± 3.2: this reversed-mask control loses fourteen points, one instantiation of the protect-the-past direction rather than a verdict on that literature. *Soft rotation* (halving rather than silencing recent winners) loses on both axes ($89.9 \pm 0.7/84.3 \pm 1.0$): all-or-none OFF periods beat turned-down gain. *Optimiser state:* masked Adam and heavy-ball SGD tie on this split ($91.5 \pm 0.4/94.1 \pm 0.1$ vs. $91.6 \pm 0.3/94.1 \pm 0.2$), and letting Adam's moments leak outside the mask adds nothing (91.6 ± 0.6); we keep the single velocity as the least optimiser state that stays confined to the mask, not for accuracy. Momentum-free SGD collapses to chance unless the replay step is re-scaled by hand, and then trails by $3.2/4.7$; a two-sided learning-rate sweep shows these comparisons are plateaus (Appendices I, H).

When replay must be rare: homeostatic bursts, not surprise. Figure 2 (right): at 12.5% of the replay events a clocked burst of 16 batches every 128 reaches 80.5 ± 6.1, no better than an even trickle at the same budget (78.5 ± 2.6); unit-level pressure triggering keeps 88.6 ± 1.6 at a third of the events, and surprise triggering fires no event (18.8) — the opposite of the offline night, whose best trigger in our sweeps is novelty-timed.

The i.i.d. price and the activity fraction. Rotation's static price is governed by the substrate's activity fraction: 1.5 points at 5% (92.6 against 94.1 rotation-free) and 1.2–1.7 at 10%, where the

Table 1: **Component ablation** (512–256, k-WTA 10%, K=1000, one replay micro-batch of 16 per waking batch; held-out split). Sequential = split-MNIST; static = i.i.d. MNIST. Headline rows six seeds, the rest three.

Variant	Sequential	i.i.d. (static)
full system: isolated replay + always-on rotation (heavy-ball SGD)	$\mathbf{91.6 \pm 0.3}$	94.1 ± 0.2
rotation gated by relative novelty, committed to bouts	91.8 ± 0.3	94.4 ± 0.3
rotation gated per batch	87.5 ± 1.6	94.7 ± 0.1
− rotation (same mask, Eq. 3, natural silence only)	88.0 ± 0.7	95.3 ± 0.2
− isolation, − rotation (unmasked interleaved replay)	85.4 ± 3.5	95.8 ± 0.3
− isolation, + rotation (rotation alone)	91.5 ± 0.3	93.7 ± 0.3
readout-only replay (hidden layers take no replay update)	41.7 ± 10.9	82.7 ± 0.3
random rotation (matched suppression count)	91.0 ± 0.8	92.1 ± 0.2
− replay (no buffer)	19.4 ± 0.0	95.1 ± 0.0
masked Adam, moments confined to the mask	91.5 ± 0.4	94.1 ± 0.1
masked Adam, moments leaking outside the mask	91.6 ± 0.6	94.0 ± 0.3
momentum 0 (pure SGD; η and replay gain re-tuned, else chance)	88.4 ± 0.5	89.4 ± 0.3
offline night instead of concurrent replay, same substrate	89.0 ± 3.5	—
the same night on its preferred narrow substrate	89.2 ± 1.7	—
mirror direction: protect the past	78.0 ± 3.2	91.7 ± 0.6
soft rotation: gain 0.5 instead of 0	89.9 ± 0.7	84.3 ± 1.0
unmasked ER at the same budget (Adam)	87.5 ± 0.8	95.1 ± 0.2
backprop (+ER sequential; plain static), dense	88.8 ± 0.3	97.4 ± 0.1

gates are neutral (Table 1); from 10 to 25% the sequential axis is flat (91.6, 91.4, 91.5) while the static axis rises (94.1, 94.4, 94.6). On feature CIFAR-10 relaxing the fraction from 5 to 25% improves both axes monotonically ($40.9 \to 42.9$ / $38.3 \to 46.4$), the local learner 1.4–1.9 points ahead of matched-sparsity BP at every level (Appendices E–F).

Transfer to CIFAR-10. On raw split CIFAR-10 the ordering transfers and widens: refractory local sleep $28.9 \pm 1.0 >$ night $25.7 \pm 1.2 >$ BP+ER $24.8 \pm 0.3 >$ unmasked ER $15.4 \pm 4.2 \approx$ no buffer 16.3 (the same on the second split: $29.7 > 26.5 > 25.8 > 15.5$). On the feature front-end the local ordering persists (refractory $42.3 \pm 0.6 >$ night $34.0 \pm 0.8 \gg$ unmasked 20.5 ± 1.0) while BP+ER reaches 42.8 ± 0.7 at its best learning rate — a 0.5-point lead. Transplanting the local network's ingredients into the BP net one at a time (each at its better of two learning rates) attributes that gap to activation sparsity (width 43.7 ± 1.0, +30% wiring 43.5 ± 0.2, +k-WTA 40.4 ± 1.5: below the local learner on the identical substrate). Converted to spiking inference (T_s=8) the default system runs at 92.7% for a proxy of 117 nJ per sample, 21× below dense rate inference (Appendix D).

5 Scaling the mechanism: self-referenced decay and depth

The last tuned constant. Every result above runs the rule with a decay λ tuned once per dataset (10^{-3} on the image benchmarks). Split CIFAR-100 exposes what that constant hides: each class receives a tenth of the supervision events while the decay acts on every synapse at every step, so at the MNIST-tuned value the hidden weights follow the pure-decay trajectory and the network lands at chance (2.4% i.i.d. against 20–23% on a ten-class subset, development-phase numbers). We replace the constant by a per-layer ratio controller,

$$\lambda_\ell = \min\Big(\lambda_{\max},\ \rho\,\mathrm{EMA}\big[u^\ell\big] / \big(\overline{|W^\ell|} + \epsilon\big)\Big), \qquad \rho = 0.25,\ \lambda_{\max} = 0.02, \tag{5}$$

where u^ℓ is the mean magnitude of the post-optimiser, *pre-decay* waking increment of layer ℓ (the decay never feeds its own drive; EMA coefficient 0.02 over unmasked waking steps, $\epsilon = 10^{-12}$), and W^ℓ, $B^{\ell-1}$ share λ_ℓ: the decay's mean magnitude $\lambda_\ell\overline{|W^\ell|}$ is held at the fraction ρ of the mean data-driven update. ρ is the only control target; $\lambda_{\max}$, the EMA coefficient and ϵ are global safeguards never tuned per dataset. Isolation and rotation decide *where* and *when* to consolidate, the controller *how strongly*; like the novelty and homeostatic signals it is self-referenced, never absolute.

One ratio target, every dataset. With the same ρ on every dataset (Table 4 in Appendix J; e.g. split-MNIST 91.6/94.1 $\to$ 90.6/93.5, CIFAR-10 28.9/28.0 $\to$ 26.2/30.1, sequential/static) the

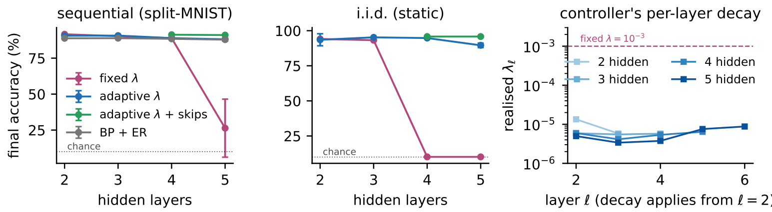

Figure 3: **Depth is the same starvation on another axis, and two mechanisms cross it.** Sequential (left) and static (middle) accuracy against hidden depth (three seeds; six at the skip system's depth five): the fixed decay without skips collapses, the controller degrades gracefully, controller + skips hold the two-layer level. Right: the controller's realised per-layer decay.

controller concedes 1–3 sequential points to the per-dataset tuned value and gains 1.2–5.3 static points on CIFAR-10, feature CIFAR-10 and both CIFAR-100 variants; on MNIST it gains 1.1 on the second held-out split (95.2 ± 0.4) but one of six seeds collapsed on the first (93.5 ± 4.3), so we claim no static gain there. On CIFAR-100 it lifts the system 2.7–4.7× over the manual rescue. The realised λ_ℓ self-differentiates as the tuning history would predict (10^{-5} on MNIST streams to 3×10^{-3} on raw CIFAR): the per-dataset ladder was a signal-to-decay ratio in disguise (Appendix J).

Depth (Figure 3). Under the tuned fixed decay without skips the static axis dies at four hidden layers ($10.1 \pm 0.2\%$, chance) and the sequential axis collapses at five (26.3 ± 20.2); under the controller every cell stays alive ($90.7 \to 88.9 \to 88.1$ sequential, $95.3 \to 94.8 \to 89.6$ static). The residual toll is architectural: ResNet-style skip synapses (Appendix B: S^ℓ carrying $a^{\ell-2}$ into each deep hidden layer, learned by the same local product with a feedback twin, replay-masked by the same rule) restore the thick error path, and the five-hidden-layer system reaches $91.0 \pm 0.3/95.8 \pm 0.3$ (six seeds) against the two-layer default's $91.6/94.1$ and the two-layer controller's $90.6/93.5$: 0.6 sequential points below the default and 1.7 above it on the static axis. The bypass alone also rescues the fixed decay ($90.4/93.5$ at depth five), so error-path attenuation is a principal cause of the collapse (the fixed decay was not re-tuned per depth; the claim is that the controller needs no re-tuning). BP+ER at the same widths is depth-insensitive ($88.9 \to 87.9$) but forgets 2× more. On CIFAR-100, by contrast, depth is a net loss that skips only half refund, while width pays (Appendix J).

6 Discussion

What this offers agents. The construction is a recipe: if an agent's representation is sparse enough that a batch leaves most units silent, and its learning rule yields a local error signal, consolidation can run in the silent degrees of freedom between the steps of the stream, with no separate offline phase — in the supervised setting studied here, with a conditional guarantee on the computation in use, at a higher total compute (11× the night's wall-clock, unoptimised). The invariance closes the channel through which consolidation would perturb the units in use; its accuracy value is 2.6 points without rotation and 0.1–0.3 with it (Table 1), so isolation's contribution is the guarantee, and at two-sample batches the collapse it prevents. We treat the biology as analogy, not mechanistic equivalence; the refractory rule mirrors the ON/OFF finding of Driessen et al. [2026] and the trigger dissociation is a testable prediction.

Limitations. All results are at MLP scale on MNIST/CIFAR variants with three to six seeds; configurations were chosen on the official test sets (the development set) and every reported number is on held-out training data (two independent splits for the headline rows); the protocol is five epochs per task (a single pass keeps the lead, Appendix I). The lead over the best offline night is 2.5 points on the first split and 0.3 on the second, where the night's seed variance vanished; the leads over BP+ER and unmasked replay hold on both. BP+ER (an external reference, not a state-of-the-art claim; DER++-style baselines are not included) keeps a 0.5-point lead on feature-based CIFAR-10; rotation keeps a static price of 1–2 points; CIFAR-100 is floor-regime; the energy figure is an arithmetic proxy. Within these limits the conclusion stands: consolidation does not require a separate offline phase, and the learner stays above the strongest offline schedule at 2.4× its replay samples and at 1.2×. Code can be found at `https://anonymous.4open.science/r/IaC-CL-with-no-offline`

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A Proofs

Proof of Proposition 1. By induction on ℓ, for every sample b. Assume $a^{\ell-1}_{\cdot,b}$ is unchanged (true for $\ell = 1$ since $a^0_{\cdot,b} = x_b$). For any unit i, $\tilde{z}^\ell_{i,b} - z^\ell_{i,b} = \sum_j \Delta^\ell_{ij} a^{\ell-1}_{j,b}$, and each term vanishes: if $j \in \mathcal{A}^{\ell-1}(X)$ then $\Delta^\ell_{ij} = 0$, otherwise $a^{\ell-1}_{j,b} = 0$ for every b. Hence $z^\ell_{\cdot,b}$, and $a^\ell_{\cdot,b}$ through (1), are unchanged; the induction closes at the last hidden layer, and at the output layer whenever the readout row satisfies the same condition. □

Proof of Proposition 2. A unit outside $\mathcal{A}^\ell(X)$ contributes $a^\ell_{i,b} = 0$ on every sample. After the update its pre-activation on sample b is $z^\ell_{i,b} + \delta_{i,b}$; under the stated margin it is still excluded by Φ_k (or rectified to zero), so $a^\ell_{i,b} = 0$ still. The sample's original winners keep their pre-activations — their synapses from active presynaptic units lie outside Δ^ℓ by the mask, and their silent-presynaptic terms vanish as in Proposition 1 — so the same k units win with the same values, every other unit's input is untouched, and the layer's activities are identical. When fewer than k units are positive on sample b, $\tau^\ell_{k,b} = 0$ and the condition reads $\sigma(z^\ell_{i,b} + \delta_{i,b}) = 0$: a zero-valued entry carries no activity whether or not Φ_k selects it. The readout row is excluded from both propositions in the default system because it is deliberately kept plastic. □

B One training step, the three channels and the skip path

Algorithm 1 lists one waking step of the default system with the state each quantity is read from. Two conventions matter for the guarantees. First, the awake sets are read from the waking forward pass, i.e. from the weights before the unmasked waking update, while the masked replay step acts on the updated weights; Propositions 1–2 hold verbatim for a mask computed on the weights being updated, and the drift reported in Section 3 is measured under the implemented order. Second, the replay micro-batch is inferred with the suppression removed, so a refractory unit may fire on replay and receive the update (Corollary 1 makes that update invisible to the waking batch for any magnitude), and a burst of n_b micro-batches reuses the mask of its waking batch. The waking step is $\eta_w = \eta\, m/256$ for a waking batch of m samples (the per-epoch drift of a batch-256 protocol), the replay step $\eta_r = 3\eta$ per micro-batch, with $\eta = 0.02$ throughout.

Algorithm 1 One waking step of the default system (rotation always on, one burst per batch).

Require: weights W_t, velocities v_t, suppression $s(t)$, buffer $\mathcal{M}$, pressures S

$a \leftarrow \mathrm{forward}(X_t;\, W_t,\, s(t))$, $\varepsilon \leftarrow \mathrm{sweep}(a, y_t)$ — Eqs. (1)–(2)
$(W, v) \leftarrow$ unmasked heavy-ball step on (W_t, v_t) with $G^\ell = \varepsilon^\ell (a^{\ell-1})^\top$, rate η_w — wake update
$\mathcal{A}^\ell \leftarrow \{i : \exists b,\ a^\ell_{i,b} \neq 0\}$ for every hidden layer; $S_i \leftarrow S_i + \bar{a}_i(X_t)$ — from the pre-update a
$M^\ell \leftarrow$ Eq. (3), unit mask $\mathbf{1}[i \notin \mathcal{A}^\ell]$, readout row and bias unmasked
$s^\ell_i(t{+}1) \leftarrow \mathbf{1}[i \in \mathcal{A}^\ell]$ — refractory rotation
write (X_t, y_t) into $\mathcal{M}$ (reservoir)
for each of the n_b replay micro-batches $(X_r, y_r) \sim \mathcal{M}$ **do**
  $a_r \leftarrow \mathrm{forward}(X_r;\, W,\, s{=}0)$, $\varepsilon_r \leftarrow \mathrm{sweep}(a_r, y_r)$ — suppression removed
  $(W, v) \leftarrow$ masked step, Eq. (4), with $G^\ell = \varepsilon^\ell_r (a^{\ell-1}_r)^\top$, rate η_r, masks M^ℓ
end for
$S_i \leftarrow 0$ for every $i \notin \mathcal{A}^\ell$ if a burst ran — discharge
return $W_{t+1} = W$, $v_{t+1} = v$, $s(t{+}1)$

The three channels. Table 2 lists the three ways a synapse enters the mask (3) and what each guarantees; the rotation of Section 3 exists to move synapses from the second row into the third.

A numerical example. On one sample with $k = 2$ and $\sigma(z) = (3, 2, 0.5, 0)$, units 3 and 4 are asleep and $\tau_k = 2$. Any replay update with $\sigma(z_3 + \delta_3) < 2$ and $\sigma(z_4 + \delta_4) < 2$ leaves the activity vector $(3, 2, 0, 0)$ unchanged, because the two winners' own inputs are not touched; if unit 4 is refractory ($s_4 = 1$) its bound is void and δ_4 may be arbitrarily large; and every synapse leaving unit 4 into the next layer may move by any amount, since its presynaptic activity is zero.

Channel	Condition	Invisible to X	Source
presynaptic silent	$j \notin \mathcal{A}^{\ell-1}(X)$	for any magnitude	Prop. 1
postsynaptic naturally asleep, active pre	$i \notin \mathcal{A}^{\ell}(X)$, $s_i^\ell = 0$	under the margin $\tau_{k,b}^\ell$	Prop. 2
postsynaptic suppressed	$s_i^\ell = 1$	for any magnitude	Cor. 1

Table 2: **The three consolidation channels** opened by the synapse mask.

Skip synapses. For $\ell \geq 3$ each deep hidden layer receives a bypass, $z^\ell = W^\ell a^{\ell-1} + S^\ell a^{\ell-2} + b^\ell$, with S^ℓ under the same Dale projection and a feedback twin $B_S^\ell \in \mathbb{R}^{n_{\ell+2} \times n_\ell}$ that carries the bypass target's burst back to its source,

$$\varepsilon^\ell \leftarrow \varepsilon^\ell + \sigma'(z^\ell) \odot \Big((B_S^\ell)^\top \big[\kappa \tanh(\varepsilon^{\ell+2}/\kappa) \odot \mathbf{1}(a^{\ell+2} > 0) \big] \Big),$$

S^ℓ and $B_S^{\ell-2}$ learned by the local product of the main text with the target layer's decay λ_ℓ, and replay-masked by the same rule two layers apart, $M_{S,ij}^\ell = \mathbf{1}[\, j \notin \mathcal{A}^{\ell-2}(X) \ \vee \ i \notin \mathcal{A}^{\ell}(X)\,]$, so that Propositions 1–2 extend unchanged.

C The price of the substrate

Table 3 traces the path from a dense backpropagation MLP to the substrate used throughout, on i.i.d. MNIST. The cumulative ladder prices the entire constraint set at 1.2 points below the dense reference for $4\times$ fewer synaptic operations per pass; the local burst-error objective itself costs nothing. Settling cannot merely be shortened ($T{=}1$ collapses); the single-phase burst sweep matches five-step settling in one pass. The Kolen–Pollack decay and even restored weight transport are accuracy-neutral at this scale — the decay earns its keep as a regulariser on harder inputs. Spiking-flavoured training fails outright: spiking is affordable at inference, not as a training-time code. On the plain substrate SGD loses 1.3 points to Adam, yet inside the consolidation loop it wins on both axes: per-synapse adaptive state is an asset while every synapse trains on every batch and a liability once updates become mask-confined.

Table 3: **Substrate ablation** on i.i.d. MNIST (256–128 hidden core, three seeds; development phase, official test set). "Syn. ops" is the fraction of the dense MLP's synaptic operations per forward pass. Upper block cumulative; lower block replaces one ingredient at its stage.

Substrate	Accuracy	Syn. ops
dense backpropagation MLP (autograd reference)	97.7 ± 0.1	1.00
local burst-error learner, unconstrained	97.8 ± 0.1	0.85
+ bounded error, Dale's law, sign-concordant feedback (no transport)	97.1 ± 0.0	0.85
+ k-WTA 10%	97.0 ± 0.2	0.80
+ relaxation $T{=}5$	97.0 ± 0.1	0.80
+ 30% distance-dependent connectivity	96.5 ± 0.2	0.24
+ burst gate and burst baseline	96.3 ± 0.2	0.24
+ single-phase burst sweep replacing relaxation (*final substrate*)	96.5 ± 0.1	0.24
random wiring instead of distance-dependent, equal density	95.4 ± 0.0	0.24
one settling step ($T{=}1$) instead of five	17.9 ± 6.8	0.80
weight transport restored (mirroring, no KP decay)	96.3 ± 0.2	0.24
Kolen–Pollack decay removed	96.3 ± 0.1	0.24
heavy-ball SGD instead of Adam ($\eta = 0.02$)	95.2 ± 0.3	0.24
k-WTA 5% / 2% instead of 10%	95.9 ± 0.1 / 94.2 ± 0.2	0.79/0.78
multiplicative burst coding	78.8 ± 1.9	0.24
training on $T_s{=}8$ spike counts	80.0 ± 0.8	0.24

D Protocol details, cost accounting and the operating frontier

The V1-like front-end consists of 256 normalised random $6\times6\times3$ patches sampled from the training images, applied with stride 2, rectified against each filter's mean response over training images, and quadrant-average-pooled (1024-D); nothing in it is learned. Split CIFAR-10 uses 32×32 luminance so that distance-dependent wiring keeps its image geometry. The buffer uses reservoir writing; the night reference uses a 256–128 core, its preferred width in our sweeps. Runs are deterministic at a fixed CPU thread count.

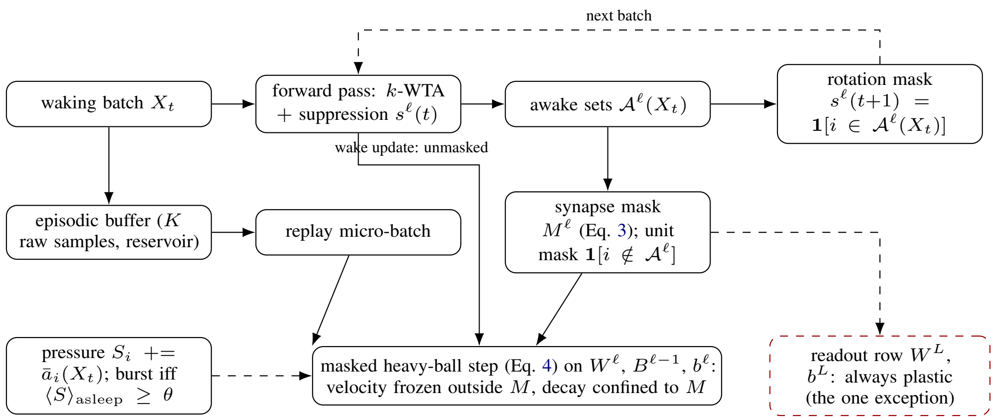

Figure 4: **Training data flow.** Each waking batch is inferred under the current suppression mask; its awake sets define the synapse and unit masks for the replay micro-batch applied in the same step and the rotation mask for the next batch. The masked step confines the velocity and the decay to the mask; the readout row is the one deliberately plastic exception. Unit-level pressure decides when replay bursts run.

Energy. Converted to spiking inference (T_s=8; thresholds calibrated on training images, evaluated on the held-out tenth), the default system runs at 92.7% (rate accuracy 94.2%) for a proxy of 117 nJ per sample against 2461 nJ for dense rate inference ($21\times$) and 526 nJ for event-driven rate inference ($4.5\times$), and is monotone in T_s (94.0% at T_s=32, 476 nJ); the rotation-free control reaches 94.2% at T_s=8 and dips to 93.9% at T_s=32 (Figure 6).

Updates and wall-clock. Samples are not the only cost: the headline schedule makes 18,760 masked updates against the night's 480 ($39\times$), and its training wall-clock on one CPU is 6.6 min against 0.6 for the night, 0.4 with no replay and 6.0 for unmasked interleaved replay — the per-batch replay step, not the sample count, is the online price of consolidating during the stream, in an unoptimised implementation whose mask construction is dense. Replay cost is reported in samples (R = replay batches $\times$ batch size). The night costs $480 \times 256 \approx 1.2 \times 10^5$ samples per run. The headline configuration (one 16-sample micro-batch per waking batch) costs $18{,}760 \times 16 \approx 3.0 \times 10^5$ ($2.4\times$ the night) for 91.6 ± 0.3. Two schedules meet the night's budget with approximately $1.22\times$: eight-sample micro-batches at every waking batch (1.5×10^5, 91.1 ± 0.4) and 16-sample micro-batches at every second one (1.5×10^5, 91.2 ± 0.2 sequential, 93.1 ± 0.2 static); pressure-triggered bursts of 16 at a third of the events reach 88.6 ± 1.6. Accuracy is bought by *frequency* of small isolated updates rather than by replayed samples. Figure 5 draws the operating frontier of the record substrate.

Baseline tuning and paired comparison. The offline-night reference was swept over replay batch count (20, 40 per night), replay batch size (16, 64, 256), learning-rate gain ($3\times$), core width (256–128 and 512–256), buffer size (200, 1000, 5000) and timing (clocked after each epoch, novelty-timed, surprise-timed), and its best sequential cell is reported; BP+ER was swept over learning rate ($3{\cdot}10^{-4}$, 10^{-3}, $3{\cdot}10^{-3}$), width and buffer size; the local learner over η (Appendix H), replay batch size, cadence and the gates of Table 1. Selection everywhere was sequential accuracy at K=1000 with the static axis reported alongside, on the official test set, which served as the development set; the numbers reported are on the held-out split (Appendix L). Pairing seeds by index on that split, the local system's margin over the narrow-substrate night is $+2.5$ points (bootstrap 95% CI $[+1.3, +4.8]$, three seed pairs), over the same-substrate night $+2.6$ ($[+0.2, +5.2]$, six pairs; the night's seed variance makes this one fragile, and on the second split it shrinks to $+0.3$), over unmasked interleaved ER $+6.3$ ($[+3.8, +9.2]$), over BP+ER $+2.8$ ($[+2.5, +3.2]$), over the silent mask $+3.6$ ($[+3.1, +4.1]$) and over rotation alone $+0.1$ ($[-0.3, +0.5]$).

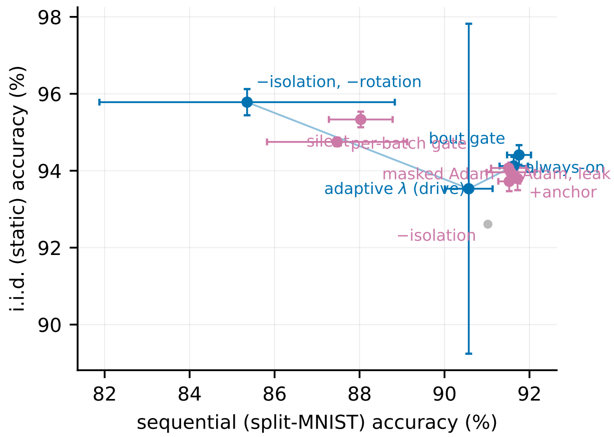

Figure 5: **Operating frontier** at the 10% activity fraction under heavy-ball SGD: split-MNIST sequential accuracy against i.i.d. static accuracy (held-out split). Blue: non-dominated configurations; purple: dominated. No single configuration wins both axes; always-on rotation is the sequential default and the adaptive decay extends the static side.

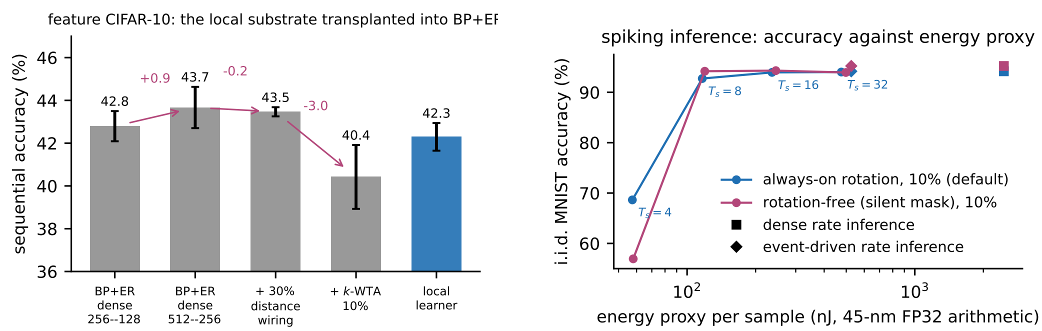

Figure 6: **Left:** the substrate decomposition on feature CIFAR-10 — transplanting the local network's ingredients into BP+ER one at a time (each cell at its better of two learning rates; six seeds except the local learner, three). **Right:** spiking inference, i.i.d. MNIST accuracy against the idealised 45-nm FP32 arithmetic-energy proxy for $T_s \in \{4, 8, 16, 32\}$, with the dense (square) and event-driven (diamond) rate references, held-out split; the default is monotone in T_s, the rotation-free control dips 0.4 at $T_s{=}32$.

E Rotation gates at the 5% activity fraction

Appendices E–H report development-phase numbers (official test set) from the earlier 5% phases and the learning-rate sweeps; they were not re-run under the held-out protocol and support mechanism orderings only. At the 5% fraction of the earlier phases, always-on rotation cost 1.3–3.0 points on i.i.d. MNIST ($95.0 \to 92.0$–93.7). Gating rotation by unit-level pressure only trades the axes (static $91.6 \to 94.0$ as sequential $88.8 \to 80.7$). The relative-novelty gate removes the price at no sequential cost on MNIST (static 95.0 ± 0.4 with rotation on 1.4–17% of batches; sequential 90.3–90.9). On split CIFAR-10 the price reverses — rotation gains $+2.5$ to $+6.5$ static points in the underfit regime — while the pure novelty gate misfires there (sequential 19.1 ± 1.1: fast/slow $\to 1$ on a stream that never becomes predictable), which the unmastered clause resolves with one setting across all four regimes (27.2 ± 0.7 / 25.3 ± 1.0 on CIFAR, 90.4 ± 0.9 / 94.3 ± 0.2 on MNIST). Those gate numbers use masked Adam; under heavy-ball SGD the per-batch gate's flicker destroys the optimiser's gain and the gate must be committed to minimum-duration bouts ($M{=}2048$ batches, loosely modelled on the consolidated bouts of the sleep–wake switch [Saper et al., 2005]; Table 1). Figure 7 shows the three regimes.

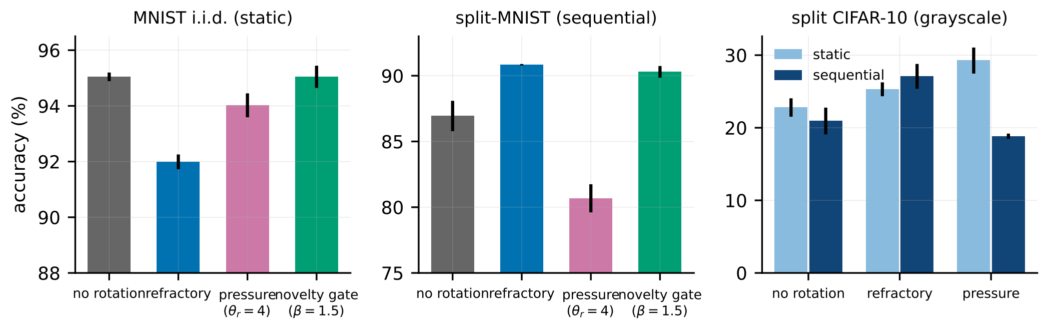

Figure 7: **Rotation at 5%: price, trade-off and gate.** Left: on i.i.d. MNIST the novelty gate restores no-rotation accuracy. Middle: on split-MNIST rotation is required and the gate keeps it. Right: on split CIFAR-10 rotation is a gain on both axes.

F Probes of the residual static price

Bout length and the OFF-side refractory. Bout length is a proper knob: $M{=}512$ gives $91.7 \pm 0.4/94.0 \pm 0.5$, recovering most of what the per-batch gate loses, while $M{=}4096$ drives the static duty to 0.94 and the refund vanishes. Completing the switch with an OFF-side refractory (triggers ignored for M_{off} batches after a bout) is a clean null: short refractories never engage on the static stream, and a long one (4096) lowers the static duty to 0.53 without moving static accuracy (94.0 ± 0.7) while monotonically eroding the sequential axis $(92.1 \rightarrow 91.0)$. The residual static price sits in the rotation of settled coalitions itself, not in how often the gate fires.

Structural exemption and synaptic anchoring. Sparing the top-q units by long-term use from ever being benched traces a smooth trade-off ($q{=}0.10$: $90.9 \pm 0.1/94.2 \pm 0.2$; $q{=}0.25$: $89.1 \pm 0.4/95.4 \pm 0.2$) that the bout gate dominates: temporal commitment beats structural exemption. Two-timescale synapses in the spirit of synaptic consolidation cascades [Benna and Fusi, 2016] — a slow anchor absorbing the fast weight at rate μ while the fast weight decays toward it at rate λ, ticking only on waking updates — do not dissolve the price (no cell exceeds the anchor-free static accuracy); weak symmetric coupling acts as a sequential stabiliser at 5% $(92.4 \pm 0.2$, variance halved) and is neutral at 10% $(92.7 \pm 0.4/94.6 \pm 0.3)$.

G Small buffers and sleep-anchored down-selection

At $K{=}200$ the night appeared to keep a 2.6-point edge; on the same 512–256/5% substrate the night reaches 84.0 ± 3.4 while noisy pressure-burst local sleep reaches 84.3 ± 1.0 and local+night 84.8 ± 0.9 — parity, with three-fold smaller variance for the wake-side scheme. The strongest tiny-buffer number remains the narrower night (86.9), so the offline window retains an advantage only in combination with narrow substrates. Replay diversity (input noise $\sigma{=}0.2$) is a wake-side lever — it lifts every local scheme $(81.8 \pm 2.9 \rightarrow 84.3 \pm 1.0)$ and hurts the night — and unmasked, rotation-free waking replay collapses (71.0 ± 1.7). Anchoring structural plasticity (prune the weakest live synapses, regrow with a distance bias) to consolidation windows, motivated by synaptic homeostasis [Tononi and Cirelli, 2014], is free on the continuous substrate $(90.9 \pm 0.2$ vs. $90.8 \pm 0.1)$, rescues the wide-sparse night $(87.2 \pm 3.8$ vs. $84.1 \pm 1.3)$, and harms the episodic-burst substrate $(84.6 \pm 0.5$ vs. $89.9 \pm 0.1)$: down-selection, like replay, needs either density or an offline window.

H Learning-rate sensitivity

At the 5% fraction, under always-on rotation, SGD holds a sequential plateau over $\eta \in [0.01, 0.02]$ $(91.9 \pm 0.5/92.0 \pm 0.6)$ and a static plateau over $[0.02, 0.05]$ (93.3–93.7), decaying gracefully outside (86.3/89.2 at $\eta{=}0.1$); masked Adam, swept over a tenfold range, has its sequential optimum at the default $(90.8 \pm 0.1$ at 10^{-3}; 88.6 at $3{\cdot}10^{-4}$, 88.9 at $3{\cdot}10^{-3})$, and no Adam setting reaches the heavy-ball joint point. Unmasked SGD is poor and unstable at every rate (84–88, seed deviations 1.8–4.1). The same two-sided sweep at the 10% record fraction (three seeds) gives the same verdict: heavy-ball SGD holds $92.2 \pm 0.7/93.3 \pm 0.2$ at η=0.01, $92.7 \pm 0.3/94.7 \pm 0.3$ at 0.02 (the joint optimum) and $91.5 \pm 0.6/94.7 \pm 0.3$ at 0.05, decaying at 0.005 ($91.3/91.3$) and destabilising at 0.1 (79.5 ± 17.3); masked Adam has its optimum at the default ($91.8 \pm 0.6/94.6 \pm 0.2$ at 10^{-3}; $89.9/89.9$ at $3{\cdot}10^{-4}$, $90.9/94.3$ at $3{\cdot}10^{-3}$). The optimiser comparison of Table 1 is made at each optimiser's best rate.

I Momentum, exactness and other negative results

Momentum. With momentum zero the default replay step (3η per micro-batch against the batch-scaled waking step $\eta\, m/256$, Appendix B) diverges or starves hidden activity at every learning rate tested ($\eta \in [0.02, 0.4]$: $\approx 10\%$, hidden dimensionality 0); the waking path alone learns normally at the matched effective step, so the collapse is the replay path's: the velocity's $\sim 10\times$ averaging of the 16-sample replay micro-batches is what heavy-ball provides. Re-scaling the replay gain (η=0.2, gain 0.3) makes pure SGD viable at $88.4 \pm 0.5/89.4 \pm 0.3$; other settings are fragile (development-phase sweep).

Exactness measurement. Under the 5% Adam-era configuration the isolated update altered the top hidden code of 1.0% of waking samples per step (against 0.32% for the record configuration); on raw CIFAR-10 (one seed) the hidden-code rate is 0.46% while the prediction rate rises to 3.6%, low-margin predictions flipping more easily through the plastic readout. The numbers in Section 3 are population rates over every replay step of a full run. They are taken under the implemented order of Algorithm 1, in which the awake sets come from the activities before the waking update; re-inferring the batch after that update and masking on the post-update awake sets lowers the hidden-code rate to 0.30% (0.17% with the readout isolated) and leaves the prediction rate (0.30%; 0.000% isolated), the margin-violation rate and the accuracy unchanged (92.2 vs. 92.0, one seed, development runs): the residual drift is the unenforced margin, not the ordering.

Replay gain. Sweeping the replay gain over $\{1/16, 1/4, 1, 3, 10\}$ (default 3; $\eta_r = 3\eta$ against a waking step of $\eta/16$) moves masked replay $67.2 \pm 3.0 \to 86.8 \pm 1.1 \to 90.8 \pm 0.1 \to 91.6 \pm 0.3 \to 74.2 \pm 32.1$ and unmasked replay without rotation $61.6 \pm 7.2 \to 81.4 \pm 4.5 \to 85.7 \pm 2.2 \to 85.4 \pm 3.5 \to 47.3 \pm 30.7$ (sequential; three seeds, six at gain 10): both peak at the default, gain 10 destabilises both (two of six seeds collapse for the masked system, four for the unmasked), and no replay gain makes the unmasked control competitive.

Single-pass streams. With one epoch per task instead of five (everything else as in the headline cells, three seeds), refractory local sleep reaches 91.8 ± 0.2 (F=5.7) against 91.6 with five epochs, BP+ER 88.7 ± 0.3, the offline night 75.1 ± 3.5 (one night per task is too few), unmasked interleaved replay 76.0 ± 20.1 and no buffer 18.1: the proposed system keeps its lead over every tested control in a single pass, while the offline night falls below BP+ER.

Falsified under matched budgets. A frozen dentate-gyrus-style sparse expansion front-end (input decorrelation does not propagate through learned k-WTA layers: input task overlap drops $0.79 \to 0.18$ yet every downstream metric worsens by 3–11 points); generative (REM) replay as a substitute for episodic samples; margin-based reverse learning; surprise-gated memory writing; multiplicative burst coding; absolute-threshold rotation gates; soft (gain-reduction) rotation; mirror-direction isolation; OFF-side gate refractories; strongly coupled synaptic anchors; utility-exempt rotation.

J The controller: composition rules, CIFAR-100 and reproducibility

Table 4: **One dataset-independent ratio target replaces per-dataset decay tuning.** Sequential / static accuracy under the tuned fixed decay (10^{-3}; 10^{-4} with $3\times$ targets as the CIFAR-100 rescue) and the controller (5) at $\rho = 0.25$ everywhere (drive form). Full system in every cell, held-out split, three seeds (six on split-MNIST); CIFAR-100 numbers are floor-regime and reported for ordering only (BP+ER 6.8 raw, 10.1 with features).

	fixed λ (tuned)		controller	
	seq.	static	seq.	static
split-MNIST	91.6 ± 0.3	94.1 ± 0.2	90.6 ± 0.6	93.5 ± 4.3
split CIFAR-10	28.9 ± 1.0	28.0 ± 1.2	26.2 ± 1.1	30.1 ± 0.2
feature CIFAR-10	42.3 ± 0.6	42.4 ± 1.1	39.1 ± 0.4	43.6 ± 0.6
split CIFAR-100	1.2 ± 0.1	1.0 ± 0.1	3.2 ± 0.8	4.1 ± 0.6
+ features	1.1 ± 0.3	1.1 ± 0.2	5.2 ± 1.6	6.4 ± 0.6

A gradient-referenced form of the controller ($\rho\,\eta\,\mathrm{EMA}[\overline{|g^\ell|}]/\overline{|W^\ell|}$) behaved equivalently in development (official test set, not re-run: split-MNIST $92.2 \pm 0.4/95.2 \pm 0.3$ against $91.7/96.1$ for the drive form there). The drive estimate must see only unamplified waking updates: inflated one-hot targets and the offline night's $3\times$-gain batches push the EMA to its clip and erase day learning (development phase: night $\times$ controller 60.2 ± 2.8 against 90.9 ± 3.2 for the night alone), whereas anchor and bout gate compose cleanly. The controller does not subsume λ's role as a capacity regulariser against overfitting on small tabular problems, information no drive ratio contains. On CIFAR-100 (ten tasks of ten classes; every method at floor, chance 1%, BP+ER 10.1 ± 0.4 with features and 6.8 ± 0.2 raw, with twice the local learner's forgetting) depth is a net loss that skips only half refund (development-phase numbers: four hidden layers $1.9/1.6 \to 3.8/3.0$ with skips against $5.7/6.0$ at two), while width at the short chain length lifts the local system to $5.8 \pm 0.9/7.7 \pm 1.1$ (1024–512): the thin-signal regime wants capacity, not depth, and the static-axis wall (6–8% under every substrate change) says the front-end is the binding constraint there. All runs reproduce bitwise at a fixed CPU thread count; a different thread count changes the reduction order, which k-WTA's discontinuity amplifies into a different trajectory.

K Isolation across replay batch sizes

Figure 8 extends the left panel of Figure 2 to two-sample replay micro-batches and shows, per batch size, the seed-paired margin of the full system over rotation with unmasked replay (held-out split; six seeds at batches 2, 4 and 16, three at 8). From 16 down to 4 the margin is 0.1 $[-0.3, +0.5]$, 0.3 $[-0.2, +0.6]$ and 0.3 $[-0.5, +1.2]$: isolation costs nothing and buys nothing measurable in mean accuracy while the rotation keeps the live coalition tolerant of unmasked replay. At batch 2 the replay gradient is too noisy for either system — the isolated one falls to 79.0 ± 3.2 (forgetting 21.5 ± 4.2 against 6.4 at batch 16), every seed between 75.5 and 84.3 — but the unmasked one bifurcates: three seeds finish at 82.5–84.8, slightly *above* their isolated counterparts ($+0.3$, $+1.8$, $+8.1$), one at 58.2 and two at chance (9.9). Leaking replay onto the live coalition is therefore not a pure loss — the surviving seeds show that the extra plasticity can help — but it opens a failure mode that the mask closes: with the current input's hidden computation held invariant, the noisiest replay degrades the system gracefully instead of destroying it. The paired mean at batch 2 ($+24$ $[-0.2, +48.5]$) describes that failure mode, not a margin, and is kept out of the ranges quoted in the main text.

L Development-phase numbers and a second held-out split

Every configuration in this paper was selected on the official test sets, which served as the development set (some four hundred sequential-axis configurations over the project); the paper reports numbers measured on a stratified tenth of the training data held out from training and never used for any decision. Table 5 places the development-phase numbers beside the reported ones and beside a second, independent held-out tenth (a different split seed) for the headline rows. Over the 87 sequential configurations run under both protocols, reported numbers sit 1.1 points below the development numbers on average and rank-correlate with them at Spearman $\rho = 0.97$; every ordering carried by

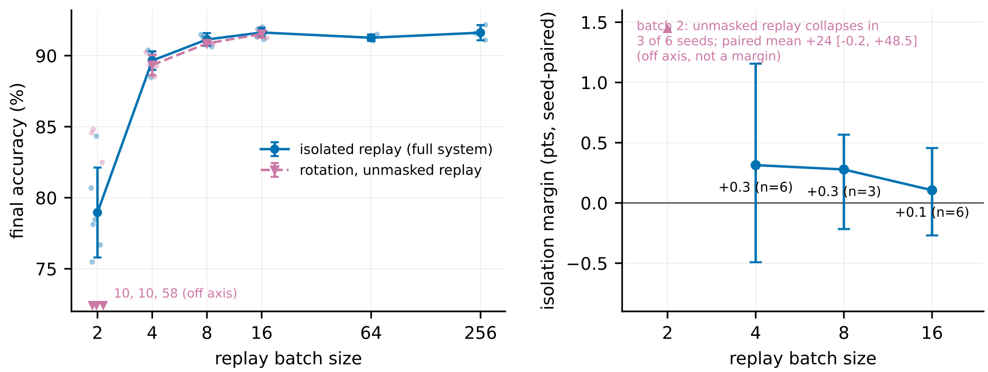

Figure 8: **Isolation across replay batch sizes** (held-out split). **Left:** split-MNIST accuracy against replay batch size for isolated replay (the full system) and for rotation with unmasked replay, every seed shown; at batch 2 no unmasked mean is drawn because three of six seeds fail (values at the axis floor). **Right:** the seed-paired margin of isolation with a 95% bootstrap interval at the batch sizes where the unmasked run does not collapse.

the paper holds on both held-out splits, with one magnitude that does not: the same-substrate night, bimodal on the first split (89.0 ± 3.5), reaches 91.5 ± 0.5 on the second and trails the full system by 0.3 there. The adaptive decay's static margin on MNIST is likewise split-dependent (93.5 ± 4.3 with one collapsed seed on the first split, 95.2 ± 0.4 on the second). Table 6 gives the per-task accuracy matrix of the headline configuration on the reported split.

Table 5: **Development-phase (official test set) numbers beside the reported held-out numbers and a second held-out split.** Sequential accuracy, with the static axis on its own row where the paper reports it; six seeds on the reported split for the headline rows, three elsewhere. Lower block: raw split CIFAR-10.

	development (test)	reported (held-out)	second split
always-on rotation (default)	92.7 ± 0.3	91.6 ± 0.3	91.8 ± 0.4
static	94.7 ± 0.3	94.1 ± 0.2	94.1 ± 0.6
bout-committed gate	92.3 ± 0.2	91.8 ± 0.3	92.0 ± 0.6
rotation alone (unmasked replay)	92.1 ± 0.5	91.5 ± 0.3	91.8 ± 0.6
static	94.7 ± 0.1	93.7 ± 0.3	93.7 ± 0.3
silent mask (no rotation)	88.8 ± 0.8	88.0 ± 0.7	88.6 ± 1.3
unmasked interleaved ER (SGD)	87.0 ± 1.1	85.4 ± 3.5	80.0 ± 14.1
BP + ER	89.3 ± 0.8	88.8 ± 0.3	88.5 ± 0.7
offline night, same substrate	90.9 ± 3.2	89.0 ± 3.5	91.5 ± 0.5
offline night, narrow substrate	90.7 ± 0.4	89.2 ± 1.7	89.3 ± 0.6
adaptive λ (drive)	91.7 ± 0.2	90.6 ± 0.6	90.8 ± 0.1
static	96.1 ± 0.2	93.5 ± 4.3	95.2 ± 0.4
five hidden layers + skips + adaptive λ	91.5 ± 0.4	91.0 ± 0.3	91.1 ± 0.3
static	96.2 ± 0.3	95.8 ± 0.3	95.7 ± 0.1
no buffer	19.6 ± 0.0	19.4 ± 0.0	19.5 ± 0.0
split CIFAR-10 (raw)			
rotation	29.5 ± 0.8	28.9 ± 1.0	29.7 ± 0.9
offline night	26.2 ± 0.6	25.7 ± 1.2	26.5 ± 0.4
BP + ER	25.1 ± 1.5	24.8 ± 0.3	25.8 ± 0.8
unmasked ER	17.0 ± 6.1	15.4 ± 4.2	15.5 ± 4.5
no buffer	16.4 ± 0.2	16.3 ± 0.1	16.6 ± 0.0

Table 6: **Per-task accuracy matrix** of the headline configuration on split-MNIST (held-out split, mean over six seeds): row = after training task i, column = accuracy on task j. Forgetting F, the mean of the diagonal minus the last row over tasks 1–4, is 6.4 ± 0.6; the drop is concentrated on tasks 2–3 and the first task is retained.

after task	T1	T2	T3	T4	T5
1	99.8				
2	98.7	97.2			
3	97.7	92.7	97.0		
4	97.5	89.6	92.9	97.5	
5	96.7	88.1	89.7	91.5	91.8